Lily Goli

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Research Interests

I am broadly interested in 3D vision, computer graphics, and robotics. My research focuses on robust 3D/4D reconstruction and camera estimation, with a recent interest in reinforcement learning for embodied AI.

Education

University of Toronto Toronto, Canada
Ph.D. (Direct Entry) in Computer Science Sept. 2021 – Expected Nov. 2026

- **GPA 4/4**

Sharif University of Technology Tehran, Iran
B.Sc. in Computer Engineering Sept. 2017 – Jun. 2021

- **GPA 19.16/20** (equivalent to major GPA of 4/4)

Research Experience

Ph.D. Graduate Research Assistant in University of Toronto Sept. 2021 - Present
Dynamic Graphics Project (DGP), Department of Computer Science Toronto, Canada
Supervisor: Professor Alec Jacobson, Professor Andrea Tagliasacchi

- Thesis: Uncertainty Estimation and Robustness in 3D Vision

Visiting Ph.D. Student at UC Berkeley Sept. 2025 - May 2026
Kanazawa AI Research group, EECS Department Remote, Berkeley, US
Supervisor: Professor Angjoo Kanazawa

- Curiosity-driven RL agent for 3D exploration

Research Intern Mar. 2026 - Present
Wayve Labs, Wayve AI Vancouver, Canada
Supervisor: Dr. Matthew Brown, Professor Andrea Tagliasacchi

- Multi-rig generative video models

Research Intern Jan. 2025 - Aug. 2025
Waabi, Sensor Simulation Group Toronto, Canada
Supervisor: Dr. Siva Manivasagam, Professor Raquel Urtasun

- Generative 3D shape completion for graphics simulation

Student Researcher Dec. 2023 - Dec. 2024
Google DeepMind, SynthX Group Remote, Mountain View, US
Supervisor: Dr. Dmitry Lagun, Professor David Fleet

- Robustness of 3D reconstruction and camera estimation for noisy scenes

Student Researcher Sept. 2021 - Present
Vector Institute, Department of Computer Science Toronto, Canada

- PhD student in computer vision and graphics

Summer Internship in Technical University of Munich (TUM) Jun. 2020 - Mar. 2021
Interdisziplinäres Forschungslabor (IFL), Computer Aided Medical Procedures (CAMP) Munich, Germany
Supervisor: Professor Nassir Navab

- Medical image segmentation given long-horizon disease development

Summer Research Program in University of British Columbia (UBC) Jun. 2019 - Sept. 2019
Robotics and Control Laboratory, Department of Electrical and Computer Engineering Vancouver, Canada
Supervisor: Professor Purang Abolmaesumi

- Volumetric representations for ultrasound images

Publications

* Equal contribution. † Equal advising

L. Goli, J. Kerr, D. Reda, A. Jacobson, A. Tagliasacchi, A. Kanazawa, "**Remember to be Curious: Episodic Context and Persistent Worlds for 3D Exploration**," arXiv preprint, 2026.

Y. Foroutan, I. Oztas, D. Rebain, A. Dundar, K.M. Yi, **L. Goli**†, A. Tagliasacchi†, "**FullCircle: Effortless 3D Reconstruction from Casual 360° Captures**," arXiv preprint, 2026.

W. Luo, **L. Goli**, S. Bahmani, F. Taubner, A. Tagliasacchi, D. Lindell "**Grow with the Flow: 4D Reconstruction of Growing Plants with Gaussian Flow Fields**," arXiv preprint, 2026.

L. Goli*, S. Sabour*, M. Matthews, M. Brubaker, D. Lagun, A. Jacobson, D. Fleet, S. Saxena†, A. Tagliasacchi†, "**RoMo: Robust Motion Segmentation Improves Structure from Motion**," International Conference on Computer Vision (ICCV), 2025.

M. Taktasheva, **L. Goli**†, A. Fiorini, Z. Li, D. Rebain, A. Tagliasacchi†, "**3D Gaussian Flats: Hybrid 2D/3D Photometric Scene Reconstruction**," Neural Information Processing Systems (NeurIPS), 2025.

J. Wang, H. Che, Y. Chen, Z. Yang, **L. Goli**, S. Manivasagam, R. Urtasun, "**Flux4D: Flow-based Unsupervised 4D Reconstruction**," Neural Information Processing Systems (NeurIPS), 2025.

A. Vora, **L. Goli**, A. Tagliasacchi, H. Zhang, "**HiT: Hierarchical Transformers for Unsupervised 3D Shape Abstraction**," International Conference on 3D Vision (3DV), 2026.

L. Goli*, S. Sabour*, M. Matthews, D. Lagun, L. Guibas, A. Jacobson, D. Fleet, A. Tagliasacchi, "**SpotLessSplats: Ignoring Distractors in 3D Gaussian Splatting**," ACM Transactions on Graphics (TOG), 2025.

L. Goli, C. Reading, S. Sellán, A. Jacobson, A. Tagliasacchi, "**Bayes' Rays: Uncertainty Quantification for Neural Radiance Fields**," Computer Vision and Pattern Recognition (CVPR), 2024. **Highlight (∼ top 10%).**

A. Shabanov, S. Govindarajan, C. Reading, **L. Goli**, D. Rebain, K.M. Yi, A. Tagliasacchi, "**BANF: Band-limited Neural Fields for Levels of Detail Reconstruction**," Computer Vision and Pattern Recognition (CVPR), 2024.

L. Goli, D. Rebain, S. Sabour, A. Garg, A. Tagliasacchi, "**nerf2nerf: Pairwise Registration of Neural Radiance Fields**," International Conference on Robotics and Automation (ICRA), 2023; Computer Vision and Pattern Recognition (CVPR) Workshop XRNeRF, 2023.

L. Goli*, S.T. Kim*, A. Khakzar, N. Navab, "**Longitudinal Quantitative Assessment of COVID-19 Infection Progression from Chest CTs**," Medical Image Computing and Computer Assisted Intervention (MICCAI), 2021.

H. Naderi, **L. Goli**, S. Kasaei, "**Generating Unrestricted Mislabeled Examples via Three Parameters**," Multimedia Tools and Applications, 2021.

H. Naderi, **L. Goli**, S. Kasaei, "**Scale Equivariant CNNs with Scale Steerable Filters**," Machine Vision and Image Processing (MVIP), 2020.

Press Coverage

Cover of the Computer Vision News: nerf2nerf with Lily Goli

Story highlight in fxguide: Stitching NeRFs: "nerf2nerf": Pairwise Registration of Neural Radiance Fields.

Bayes' Rays video presentation during Prime Minister Mark Carney's visit to the Vector Institute.

Honors and Awards

Ontario Graduate Scholarship (OGS) — $15000, 2025-2026

NeurIPS Top Reviewer — 2025

Robert E. Lansdale/Okino Computer Graphics Graduate Fellowship — $2000, 2025

Voted Top 10 Fast-Forward Presentation, SIGGRAPH — 2025

Robert E. Lansdale/Okino Computer Graphics Graduate Fellowship — $2000, 2024

Admitted and granted scholarship for International Computer Vision Summer School (ICVSS) — 2024

Awarded admission to the Master's program at Sharif University based on academic merit — 2021

Ranked 38th in the National Universities Entrance Exam for Bachelor of Science — Aug. 2017
among more than 150,000 participants.

National Elite Foundation Fellowship — 2017

Invited Talks

- **nerf2nerf** — Google Geo, Mountain View, USA 2023
  Host: Veselin Dikov
- **Bayes' Rays** —
  Google, Mountain View, USA 2024
  Host: Mark Matthews

  International Conference on 3D Vision (3DV), Singapore 2025
  Highlight Papers Section

  Kyoto Vision Workshop, Kyoto, Japan 2025
  Host: Prof. Ko Nishino
- **SpotLessSplats** —
  3D Gaussian Splatting Meetup (Online) 2024
  Host: Mike Caronna

  Google Reading Group, Mountain View, USA 2024
  Host: Prof. Leonidas Guibas
- **Uncertainty & Robustness in 3D Reconstruction** —
  ETH Zurich, Switzerland 2024
  Hosts: Xu Haofei, Prof. Marc Pollefeys

  Google Zurich, Switzerland 2024
  Hosts: Dr. Michael Oechsle, Dr. Federico Tombari

  University of California, Berkeley, USA 2025
  Host: Prof. Angjoo Kanazawa

  École des Ponts ParisTech, Paris, France 2025
  Host: Prof. Vincent Lepetit

  York University, Toronto, Canada 2025
  Host: Prof. Kosta Derpanis
  Kyoto University, Kyoto, Japan 2025
  Host: Prof. Ko Nishino

Mentoring and Student Supervision

- **Ahan Shabanov** (PhD. student, Simon Fraser University) Fall 2023
  Topic: Level of Detail representation in Neural Fields
- **Maria Taktasheva** (PhD. student, Simon Fraser University) Winter 2024 – Present
  Topic: Planar surface representation in 3D models
- **Aditya Vora** (PhD. student, Simon Fraser University) Summer 2025
  Topic: Hierarchical Shape Abstraction given 3D geometry
- **Weihan Luo** (Undergraduate student, University of Toronto) Spring 2025 - Present
  Topic: 4D reconstruction of plant growth
- **Lulu Wei** (MSc. student, University of Toronto) Fall 2025 - Spring 2026
  Topic: Robust 4D reconstruction
- **Ipek Oztas** (Visiting student, Simon Fraser University) Fall 2025 - Spring 2026
  Topic: Robust reconstruction with 360° captures
- **Yalda Foroutan** (PhD. student, Simon Fraser University) Fall 2025 - Present
  Topic: Robust reconstruction with 360° captures

Teaching Experience

Teaching Assistant at University of Toronto, Toronto, Canada Fall 2021 - Present

Foundations of Computer Science, Data Science I, Intro to Image Understanding, Intro to Computer Graphics, Introduction to Machine Learning

Teaching Assistant at Sharif University of Technology, Tehran, Iran — Fall 2019 - Spring 2021
Linear Algebra, Probability and Statistics, Artificial Intelligence

Technical Skills

Programming Languages: Python (Proficient), C (Proficient), Java (Proficient), CUDA, MATLAB

Frameworks and Tools: Viser, Gsplat, NerfStudio, Codex, Claude Code, PyTorch, JAX, Blender

Academic Service

Reviewer at CVPR, ECCV, ICCV, NeurIPS, ICLR, SIGGRAPH, SIGGRAPH Asia, TOG, TPAMI, ICRA, IROS, RA-L, AAAI.

Organizing Geometric Intelligence workshop (GeoInt) at ECCV — Oct. 2026

Organizing Generative Scene Completion workshop (GeoFreeNVS) at CVPR — Jun. 2026

Organizing 4D Vision workshop (4DV) at CVPR — Jun. 2026

Organizing Generative Scene Completion workshop (SceneComp) at ICCV — Oct. 2025

Volunteer Tutor at Her Code Camp — Sept. 2025

Organizing team for DGP high school outreach, DGP Academy (news) — Feb. 2024

Organizing 3D Vision Reading Group at University of Toronto — Oct. 2023 - Jan. 2024

Graduate Applications Triager at University of Toronto — Dec. 2023